# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE UNIT TEST SET C

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** Which quantity remains the same everywhere? 1

**(a)** force of friction **(b)** gravitational pull
**(c)** mass **(d)** weight

**2.** An object thrown upwards slows down because of: 1

**(a)** its own colour **(b)** gravitational force
**(c)** air only **(d)** magnetic force

**3.** The cyclone Amphan of 2020 had peak winds of about: 1

**(a)** 270 km/h **(b)** 27 km/h **(c)** 70 km/h **(d)** 500 km/h

**4.** A sea breeze blows: 1

**(a)** from land to sea during the day **(b)** only in winter
**(c)** from sea to land at night **(d)** from sea to land during the day

**5.** A gas takes the volume of its container because: 1

**(a)** its particles are fixed
**(b)** it has no particles
**(c)** it has strong forces
**(d)** its particles move freely in all directions

**6.** A drop of ink spreads in water because: 1

**(a)** particles of ink and water mix **(b)** ink melts
**(c)** water freezes **(d)** water evaporates

**7.** The melting point of iron is about: 1

**(a)** 1538 °C **(b)** 133 °C **(c)** 0 °C **(d)** 100 °C

**8.** Pure substances can be: 1

**(a)** only elements **(b)** elements or compounds
**(c)** only alloys **(d)** only mixtures

**9.** Which is NOT a pure substance? 1

**(a)** Baking soda **(b)** Oxygen **(c)** Fruit juice **(d)** Sugar

**10.** Magnesium burns in oxygen to form: 1

**(a)** magnesium only **(b)** magnesium chloride
**(c)** magnesium oxide **(d)** magnesium sulfide

**11.** In a salt–water solution, the solute is: 1

**(a)** salt **(b)** air **(c)** container **(d)** water

**12.** The SI unit of density is: 1

**(a)** $N/m^2$ **(b)** $kg/m^3$ **(c)** $m^3$ **(d)** kg

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 10–15 words each.*

**13.** State two properties of magnetic force. 2

**14.** Why do broad straps on a school bag help the shoulders? 2

**15.** What is meant by the particulate nature of matter? 2

**16.** Give two examples each of uniform and non-uniform mixtures. 2

**17.** Define solubility. 2

## SECTION — C (Short Answer Type) 6 × 3 = 18

*Answer in about 30–40 words each.*

**18.** Describe the vertical motion of a ball thrown upwards. 3

**19.** Explain with reasons why the base of a dam is thicker. 3

**20.** Draw the arrangement of particles in aluminium foil, glycerin and methane gas. 3

**21.** Distinguish between saturated, unsaturated and dilute solutions. 3

**22.** Write the effects of force with one example each: start motion, change shape, change direction. 3

OR

A spring balance has range 0–10 N with 1 N divided into 10 divisions. Find the least count. A load stretches it to 3 divisions after 4 N. What is the reading? 3

**23.** Calculate the force exerted by the atmosphere on 15 cm × 15 cm area, as given in the book, and state its equal mass. 3

OR

Write three safety measures during a thunderstorm. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 80–100 words each.*

**24.** Explain liquid pressure. Write its properties and its applications in dams and water tanks. 5

OR

Explain the formation of a storm and a thunderstorm step by step. 5

**25.** Differentiate mixtures and compounds in a table of five points. 5

OR

Describe the activity of heating iron and sulfur. State observations, conclusions and the word equation. 5

## END OF QUESTION PAPER Total: 50 Marks